

Chemical Bond

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Chemical Bond

The attraction force that acts between two or more atoms to hold them together as a stable molecule is called chemical bond.

There are three types different types of chemical bonds recognised by chemists

- (1) Ionic or Electrovalent bond
- (2) Covalent bond
- (3) Coordinate bond

Principles of Electronic Theory of Chemical Bond

- (1) Chemical combination between atoms of the same or different elements takes place due to the tendency by the outermost electron groups to attain the stable arrangement of inert gases
- (2) The attainment of inert gas electron configuration may take place by complete transference of electrons from one atom to another. The resulting electrically charged atoms (or ions) are held together by electrostatic force of attraction. The chemical bond is known as ionic bond.
- (3) The attainment of inert gas configuration may occur by sharing of electrons (in pairs) between two atoms. The chemical bond so formed is called covalent bond.
- (4) The attainment of electron groupings of inert gases may also happen by both transference and sharing of electrons between atoms in pairs. The bond so formed is known as coordination bond.

Ionic Bond

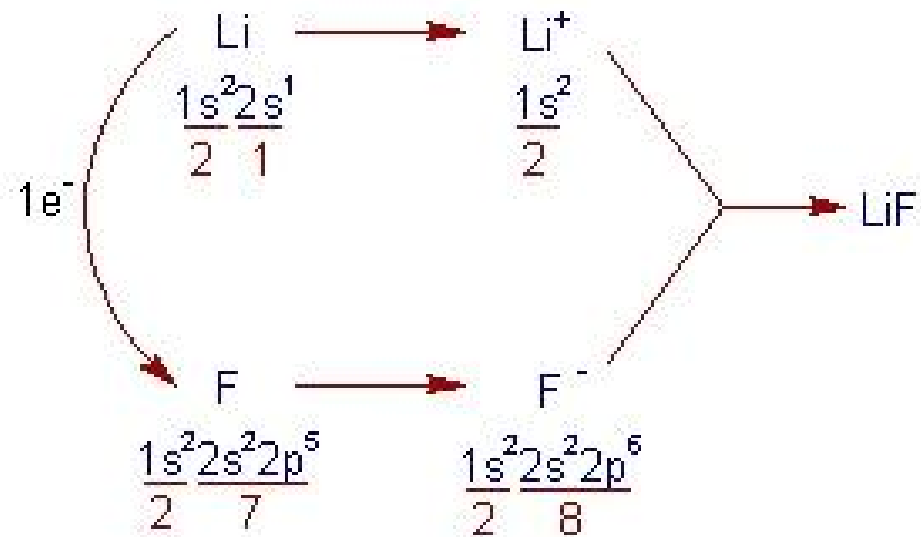
The ionic bond is the electrostatic force of attraction between two oppositely charged ions i.e., a positively charged cation and a negatively charged anion. It is formed due to complete transfer of electrons from one atom to another atom.

Formation of ionic bond

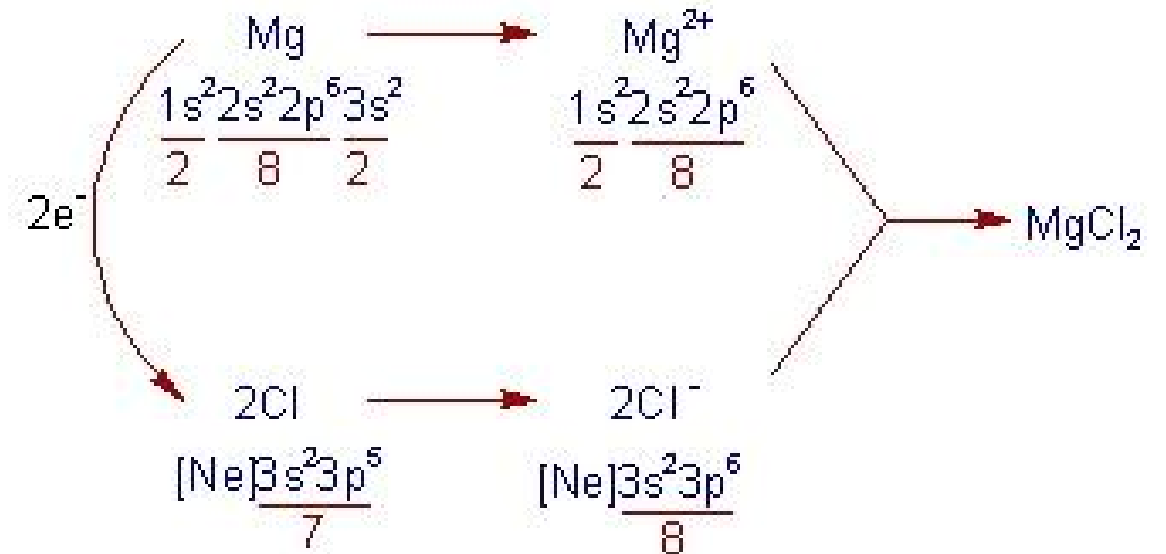
- a) An electropositive atom (metal) loses electron(s) to form a positively charged ion called as cation.
- b) An electronegative atom accepts the electron(s) to form a negatively charged ion, otherwise known as anion.
- c) Thus formed oppositely charged ions come closer to each other due to electrostatic force of attraction and get stability.

An ionic bond is formed when the electronegativity difference between two bonding atoms is greater than 1.7 on Pauling's scale.

1) LiF (Lithium fluoride):



2) MgCl_2 (Magnesium chloride):



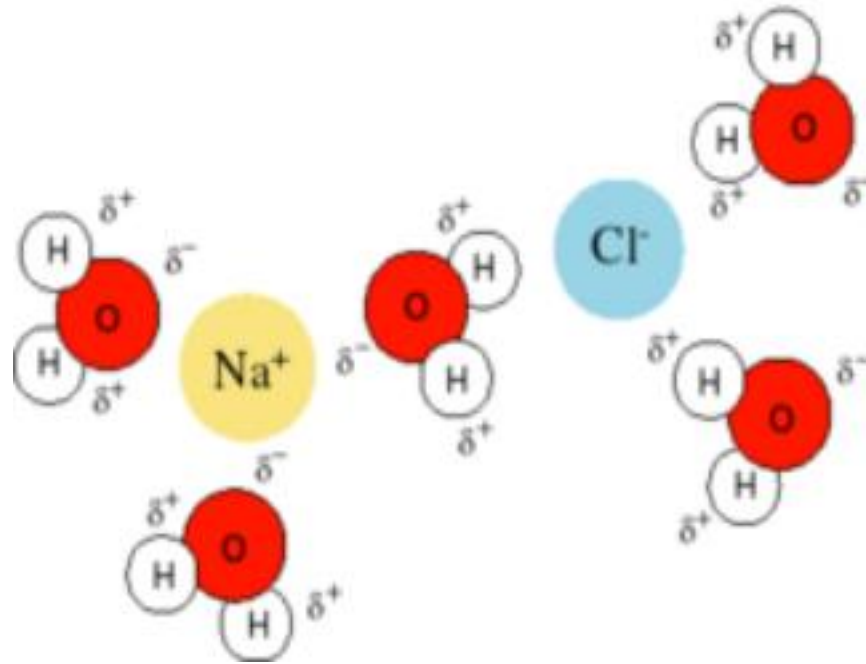
Characteristics of Ionic Compounds

- The cations and anions in these compounds are held together there by electrostatic lines of forces.
- These compounds are polar, soluble in polar solvent such as H_2O , NH_3 etc., but insoluble in non polar solvents such as CCl_4 , C_6H_6 etc.

An ionic compound consists of two oppositely charged ions. Water, on the other hand is a polar solvent; the electronegativity difference between oxygen and hydrogen is high which is why water has a positive pole of H and a negative pole O (water is H_2O)

When an ionic compound is put in water, the negative ion, or the anion, attracts the positive H pole around it, and the positive ion, or the cation, attracts the negative O pole around it. This results in the formation of a unique arrangement called the hydration.

Hydration releases energy, which is known as the hydration energy. Every ionic compound has an energy holding the lattice structure of the compound, known as lattice energy. If the hydration energy of an ionic compound exceeds its lattice energy, the lattice is broken and the ions in the compound separate, causing the compound to dissolve.

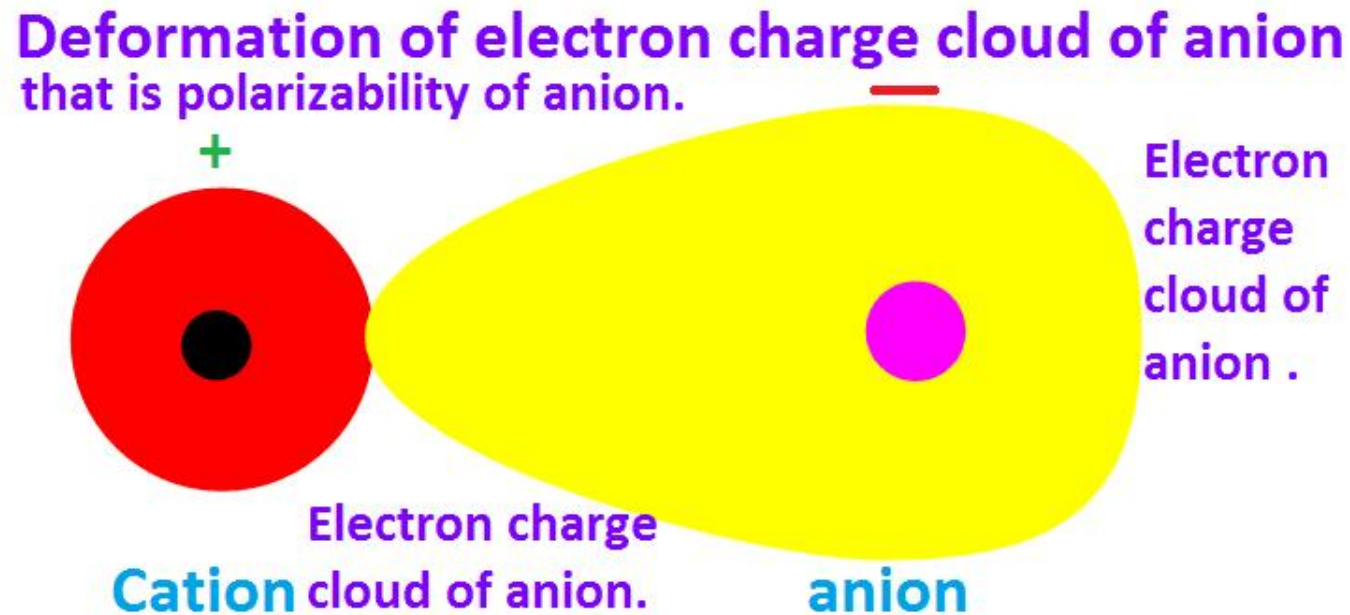


- Solutions of these compounds are good conductors of electricity.
- They possess high melting points and boiling points .

Fajan's Rule

Covalent Character of Ionic Compound:

Polarisation of an ionic bond means the distortion of the electron cloud of an anion towards a cation.



Rule-1: The smaller the cation or the larger the anion, the greater is the covalent nature of an ionic compound.

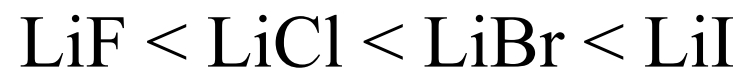
We know that the size of cations increases from top to bottom in a given group of periodic table. Therefore the cation size of alkali metals also increases as shown below.



Therefore the covalent nature of alkali metal halides containing the same halide ion decreases from Lithium halide to Cesium halide according to Fazan's rule-1. For example, the decreasing order of covalent nature of alkali metal chlorides is:



For example the order of covalent character for Lithium halides with different halide ions is as shown below.



In the above case, the size of anion increases from fluoride to iodide ion and hence the covalent nature increases

Rule-2: The greater the charge on either cation or anion, the greater is the covalent nature of an ionic compound.

The covalent nature of chlorides of Na, Mg & Al is in the following order.



It is due to increase in the positive charge on cations i.e. $\text{Na}^+ < \text{Mg}^{2+} < \text{Al}^{3+}$.

O^{2-} is more polarised than F^- because F^- has only one negative charge while O^{2-} has two negative charges.

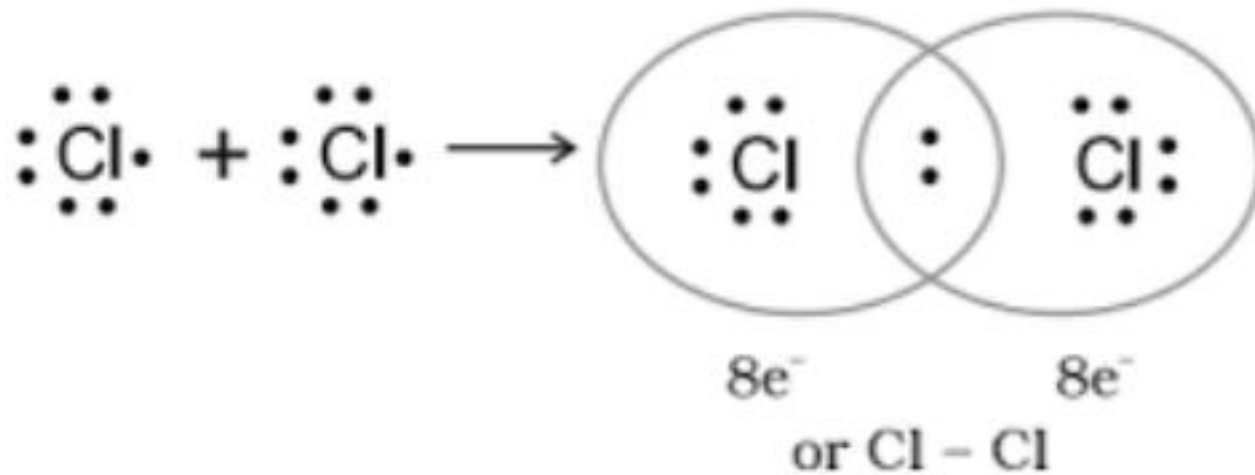


Rule 3: A cation with 18 electrons in its outermost shell has greater polarising power to polarise an anion than a cation with 8 electrons in its outermost shell even both the cations have the same size and charge.

Example : Melting point of CuCl (442°C), NaCl (800°C) [Size of $\text{Na}^{+}=0.96^{\circ}\text{\AA}$, $\text{Cu}^{+}=0.95^{\circ}\text{\AA}$]

Covalent Bond

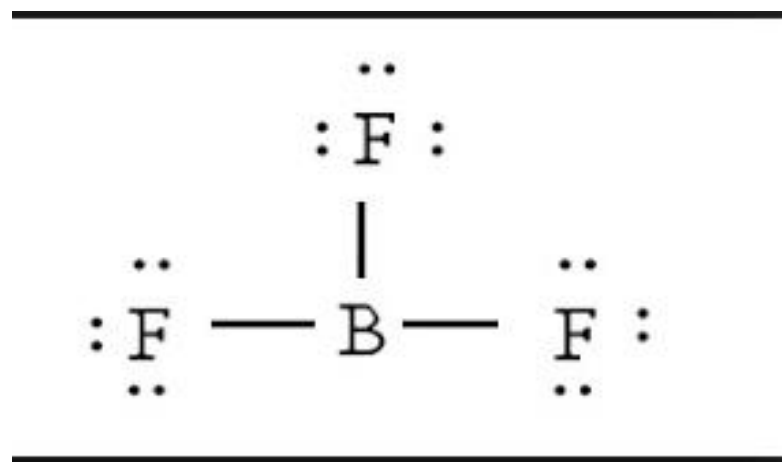
The sharing of pair of electrons between two atoms is called covalent bond.



Covalent bond between two Cl atoms

The Octet Rule : When two atoms form covalent bonds, they attain an inert gas configuration with an octet of electrons. This is known as the octet rule.

Incomplete octet: There are few molecules in which one of the atoms has not completed the required number of eight electrons. Thus in the case of BF_3 , the Boron atom has the electronic configuration $1s^2 2s^2 2p^1$. It may be expected that Boron would give up its three outermost electrons to give B^{3+} ; but these electrons are tightly held. Instead Boron shares three electrons with other atoms, say F, to form covalent BF_3 . But the boron atom does not conform to the stable configuration since it has only 6 electrons in BF_3 . [BeCl_2 , NO etc.]

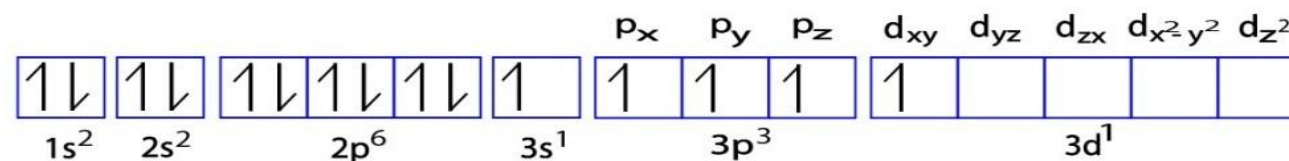


Expansion of octet: Electronic configuration for molecules $\text{PCl}_5, \text{SF}_6$ do not correspond with Lewis theory and apparently sharing of electrons in pairs involves more than 8 electrons (10 in PCl_5 and 12 in SF_6)

Q. PCl_5 exists but NCl_5 does not-Explain

The electronic configuration of P is $1s^2 2s^2 2p^6 3s^2 3p^3$

P has a vacant 3d-subshell. So, when Cl atoms come to bond to the P atom, one of the paired electrons in 3s of P get excited to 3d.



After this the P atom will undergo sp^3d hybridization and it will have 5 unpaired electrons in its valence shell. So P will be able to form 5 bonds with 5 Cl atoms each. So PCl_5 exists.

On the other hand, N does not have vacant d orbital and hence cannot promote the paired electrons in 3s. So the maximum number of unpaired electrons that N can have in its valence shell is 3 (the electrons in the 2p subshell). So NCl_5 does not exist but NCl_3 does.

Characteristics of covalent bond

- Soluble in organic solvents
- Their melting and boiling points are low
- These are generally soft, easily fusible and volatile.

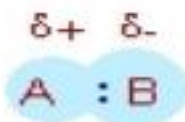
Polarity of Covalent Bond

The bond pair is equally shared in between two atoms when the electronegativity difference between them is zero or nearer to zero. In this case, neither of the atoms gets excess of electron density and hence carry no charge. This is called non polar covalent bond.



non polar covalent bond

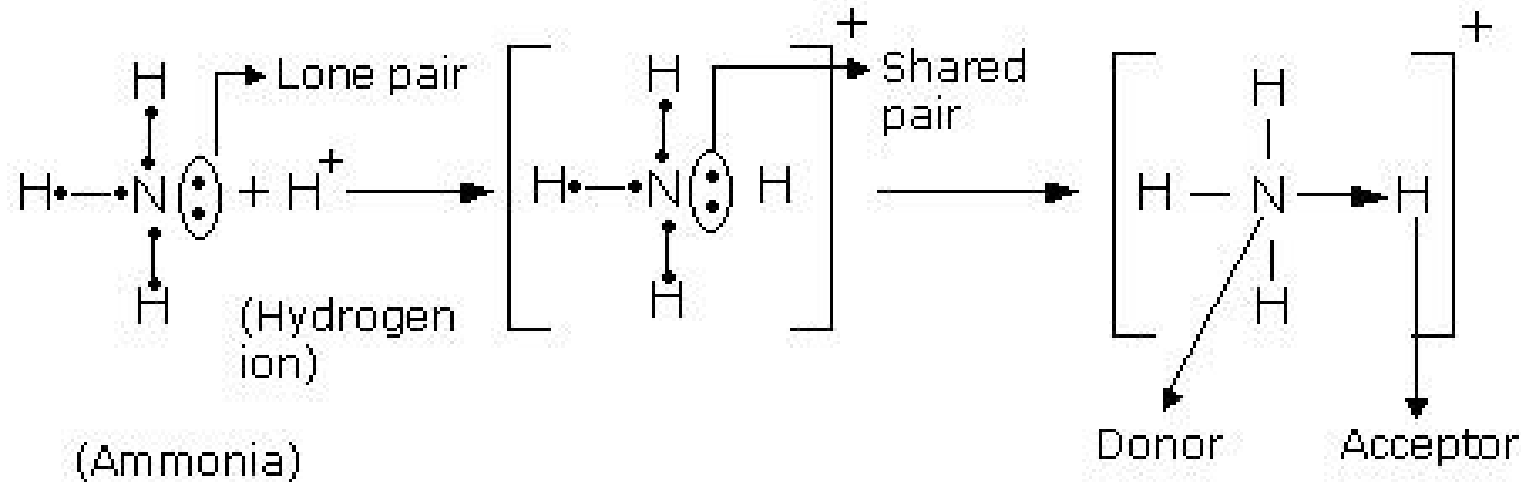
However, when there is a considerable difference in the electronegativity, the bond pair is no longer shared equally between the atoms. It is shifted slightly towards the atom with higher electronegativity by creating partial negative charge (represented by δ^-) over it. Whereas, the atom with less electronegativity gets partial positive charge (represented by δ^+). This type of bond is also referred to as polar covalent bond.



polar covalent bond
Note: B is more electronegative

Coordinate or Dative Bond

A covalent bond in which both electrons of the shared pair come from one of the two atoms (or ions) is called coordinate bond.



Characteristics of coordinate bond

- They do not ionise in water and are poor conductors of electricity.
- They are very sparingly soluble in water but dissolve in inorganic solvents.
- Since a coordinate linkage is semi polar, the coordinate compounds possess melting and boiling points which are higher than those of purely covalent bond but lower than ionic compounds.

VSEPR(Valence-Shell Electron Pair Repulsion) Theory

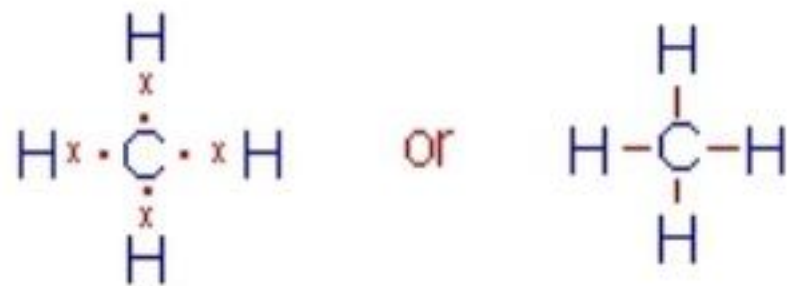
Rule 1: If the central atom of a molecule is surrounded only by bonding electron pairs (bp's) and not by lone pairs(lp's) the geometry of the molecule will be regular.

It will be BeCl_2 (bp's = 2, linear), BCl_3 (bp's=3, triangular planar), CH_4 (bp's=4, tetrahedral), PCl_5 (bp's=5, trigonal bipyramidal) and SF_6 (bp's=6, Octahedral). In all these molecules lp's=0

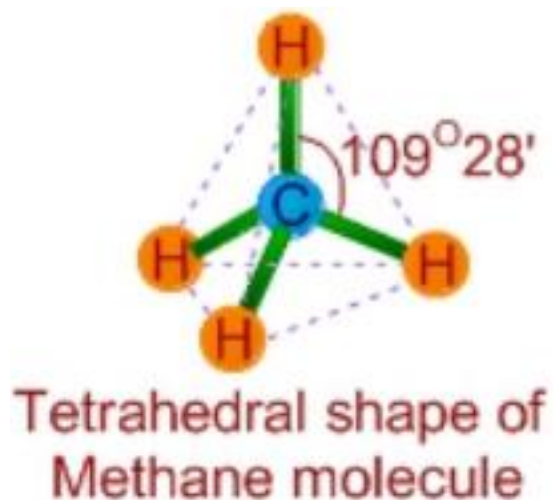
Rule 2: When the central atom in a molecule is surrounded by both bp's and lp's, the molecule does not have a regular shape. As a result the bond angle is decreased.

The order of repulsion is $\text{lp-lp} \gg \text{lp-bp} > \text{bp-bp}$

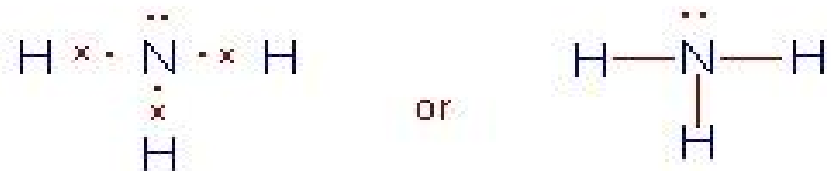
(1)Methane (CH₄): The Lewis structure of methane molecule is:



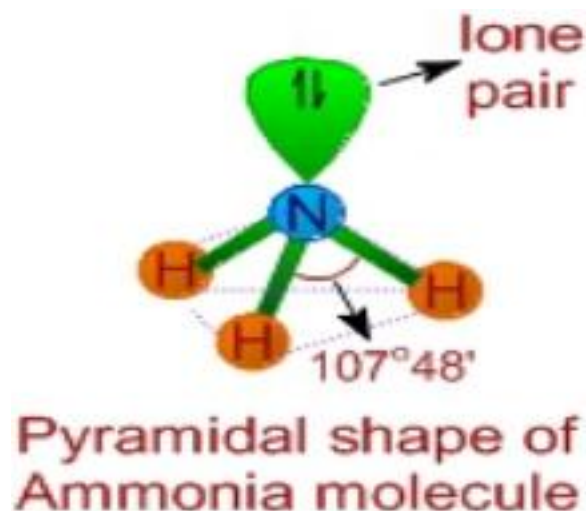
There are 4 bond pairs around the central carbon atom in its valence shell. Hence it has tetrahedral shape with 109°28' of bond angles.



(2) Ammonia (NH₃): The Lewis structure of ammonia indicates there are three bond pairs and one lone pair around the central nitrogen atom.

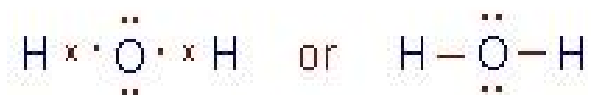


It has 3 bp's and 1 lp's electrons. However, its shape is pyramidal with a lone pair on nitrogen atom. The bond angle is decreased from 109°28' to 107°48' due to repulsion caused by lone pair on the bond pairs.

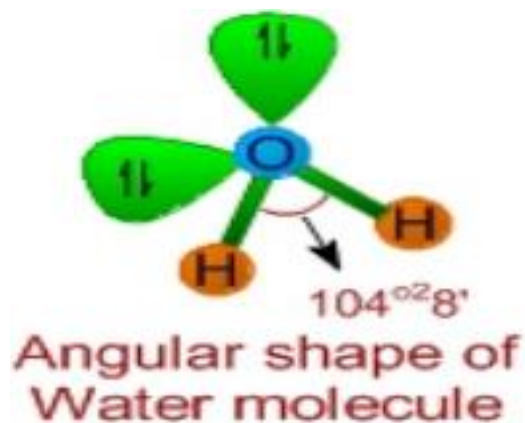


3) Water (H₂O):

There are two bond pairs and two lone pairs in the valence shell of oxygen. Hence its structure is based on tetrahedral geometry. However its shape is angular with two lone pairs on oxygen.



The bond angle is decreased to 104°28' due to repulsions caused by lone pairs on bond pairs. It can be noted that the bond angle decreases with increase in the number of lone pairs on the central atom



VBT (Valence Bond Theory)



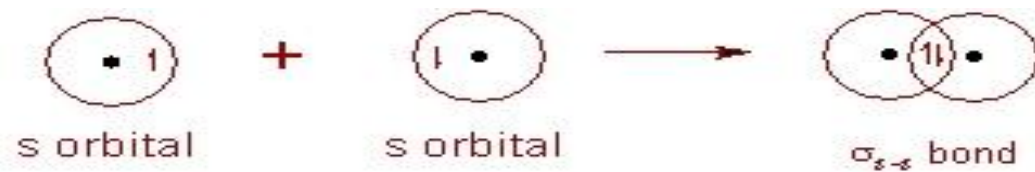
The valence bond theory was proposed by Heitler and London to explain the formation of covalent bond quantitatively using quantum mechanics. Later on, Linus Pauling improved this theory by introducing the concept of hybridization.

- The main postulates of this theory are as follows:
- A covalent bond is formed by the overlapping of two half filled valence atomic orbitals of two different atoms.
- The electrons in the overlapping orbitals get paired and confined between the nuclei of two atoms.
- The electron density between two bonded atoms increases due to overlapping. This confers stability to the molecule.
- Greater the extent of overlapping, stronger is the bond formed.
- The direction of the covalent bond is along the region of overlapping of the atomic orbitals i.e., covalent bond is directional

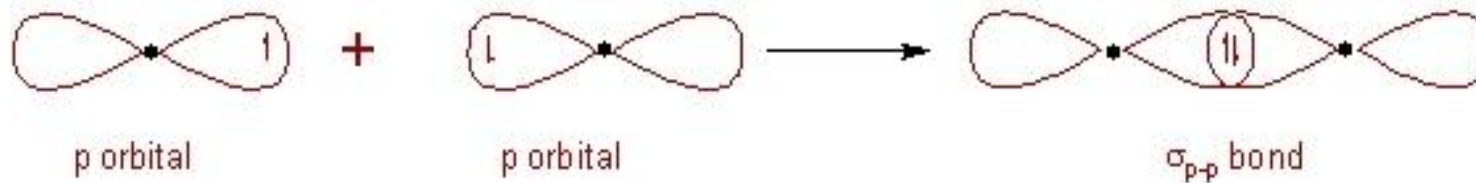
There are two types of covalent bonds based on the pattern of overlapping as follows:

(i) σ -bond: The covalent bond formed due to overlapping of atomic orbital along the inter nucleus axis is called σ -bond. It is a stronger bond and cylindrically symmetrical. Depending on the types of orbitals overlapping, the σ -bond is divided into following types:

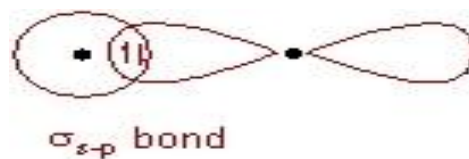
σ_{s-s} bond:



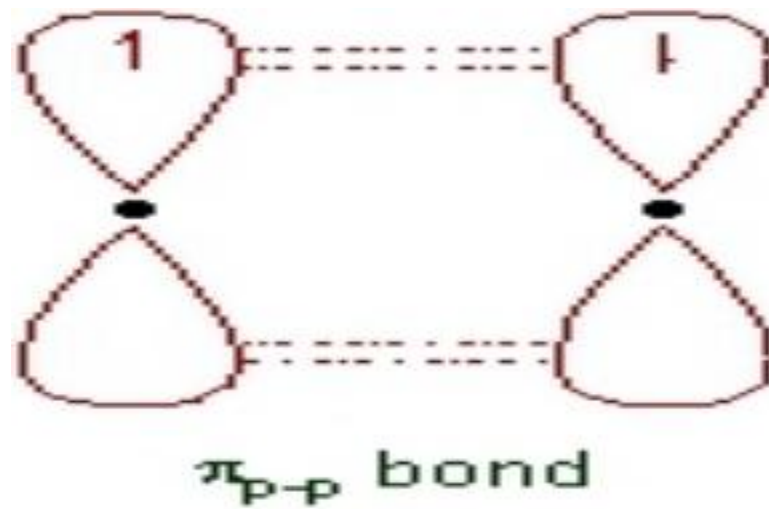
σ_{p-p} bond:



σ_{s-p} bond:



(ii) π -bond: The covalent bond formed by sidewise overlapping of atomic orbitals is called π - bond. In this bond, the electron density is present above and below the inter nuclear axis. It is relatively a weaker bond since the electrons are not strongly attracted by the nuclei of bonding atoms.



Hybridization

The intermixing of two or more pure atomic orbitals of an atom with almost same energy to give same number of identical and degenerate new type of orbitals is known as hybridization. The new orbitals formed are also known as hybrid orbitals.

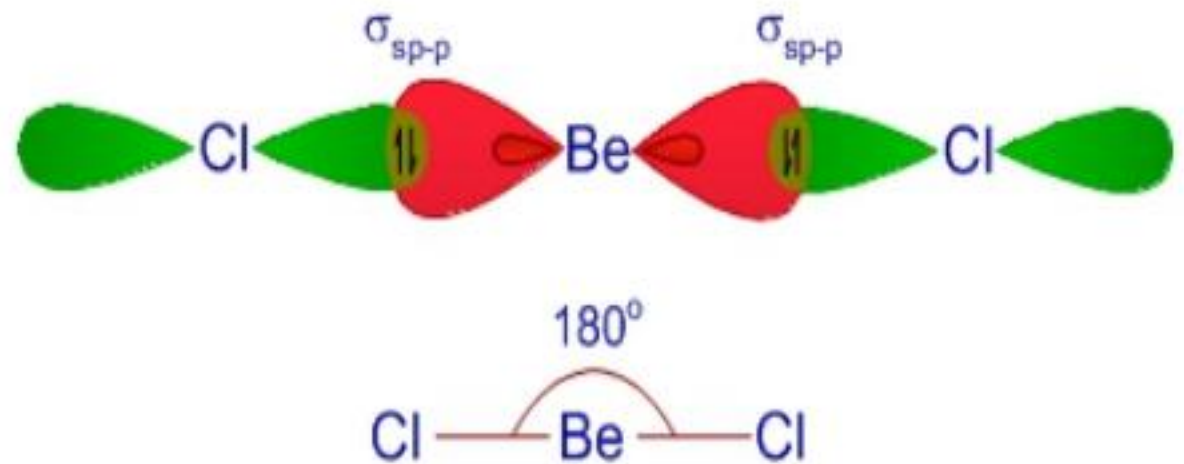
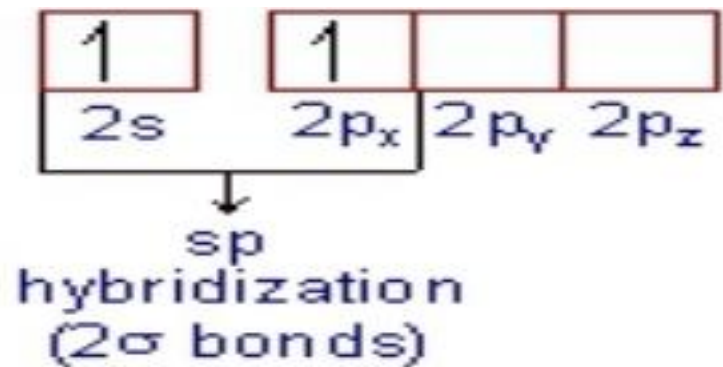
S.No.	Hybridisation	Shape	Example
1	sp	linear	CO_2 , BF_2 , C_2H_2
2	sp^2	trigonal planar	BCl_3 , C_2H_4
3	sp^3	tetrahedral	CH_4 , C_2H_6
4	dsp^2	square planar	PtCl_4^{2-}
5	sp^3d	trigonal bipyramidal	PCl_5
6	sp^3d^2	octahedral	SF_6

➤ 1) Beryllium Chloride (BeCl_2)

The electronic configuration of 'Be' in ground state is $1s^2 2s^2$

Thus in the excited state, the electronic configuration of Be is $1s^2 2s^1 2p_x^1$

In the excited state, the beryllium atom undergoes 'sp' hybridization by mixing a 2s and one 2p orbitals. Thus two half filled 'sp' hybrid orbitals are formed, which are arranged linearly

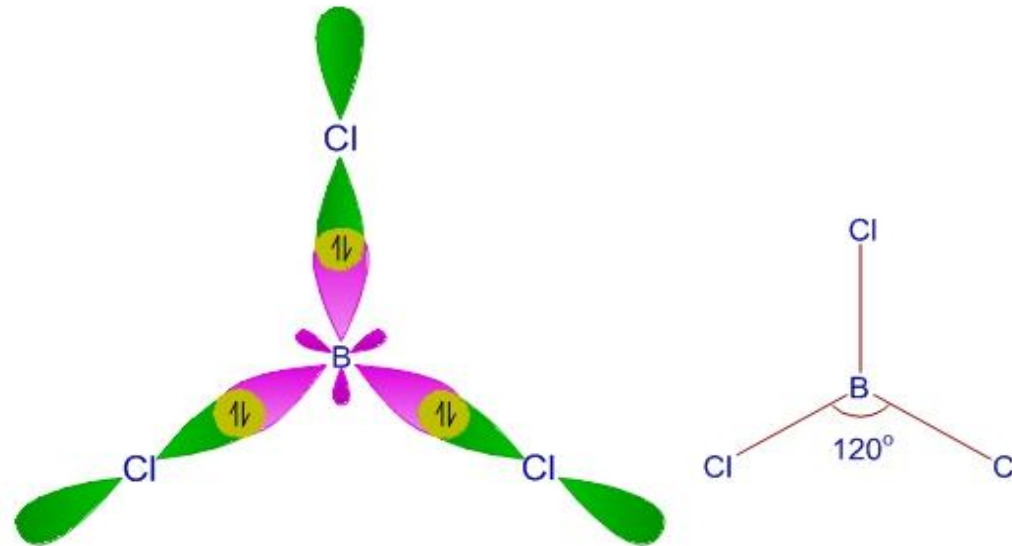
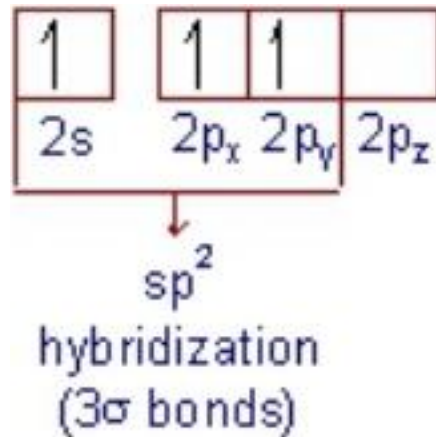


➤ 2) Boron trichloride (BCl_3)

The electronic configuration of 'B' in ground state is $1s^2 2s^2 2p^1$

Thus Boron atom gets electronic configuration: $1s^2 2s^1 2p_x^1 2p_y^1$.

In the excited state, Boron undergoes sp^2 hybridization by using a 2s and two 2p orbitals to give three half filled sp^2 hybrid orbitals which are oriented in trigonal planar symmetry



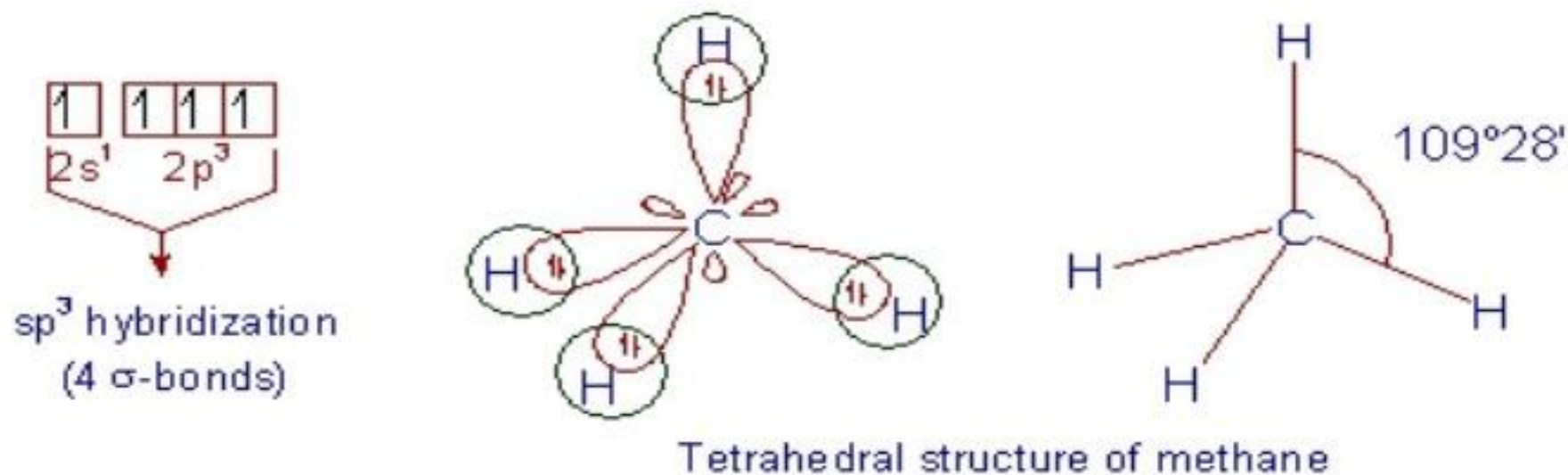
Trigonal planar structure of BCl_3

➤ 3) Methane (CH_4)

The electronic configuration of 'C' in ground state is $1s^2 2s^2 2p^2$

Thus in the excited state, the electronic configuration of C is $1s^2 2s^1 2p_x^1 2p_y^1 2p_z^1$

During the formation of methane molecule, the carbon atom undergoes sp^3 hybridization in the excited state by mixing one '2s' and three 2p orbitals to furnish four half filled sp^3 hybrid orbitals, which are oriented in tetrahedral symmetry in space around the carbon atom

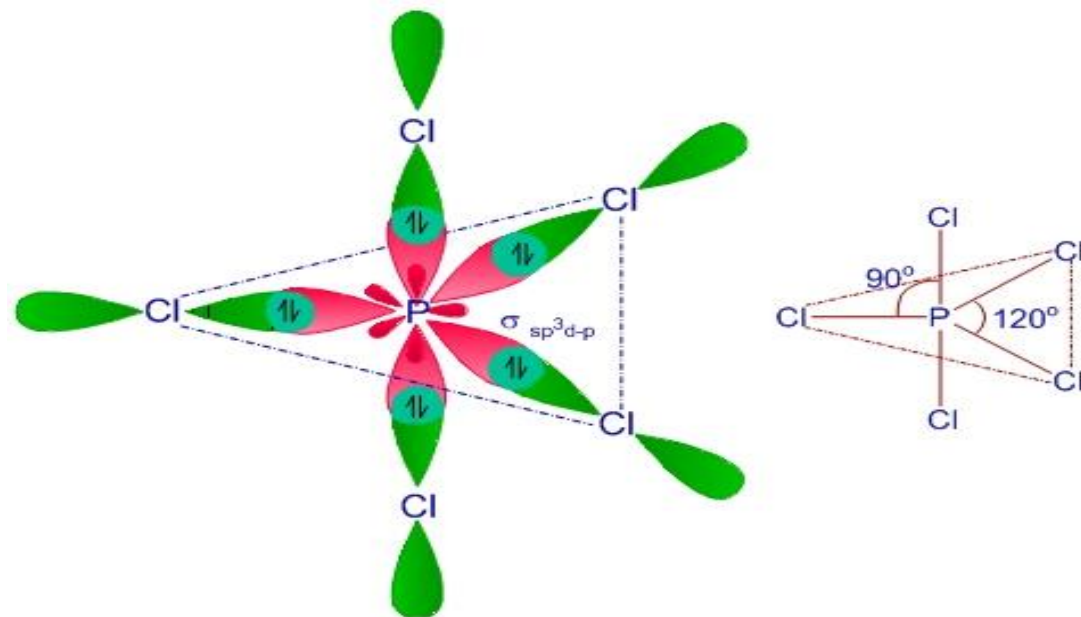
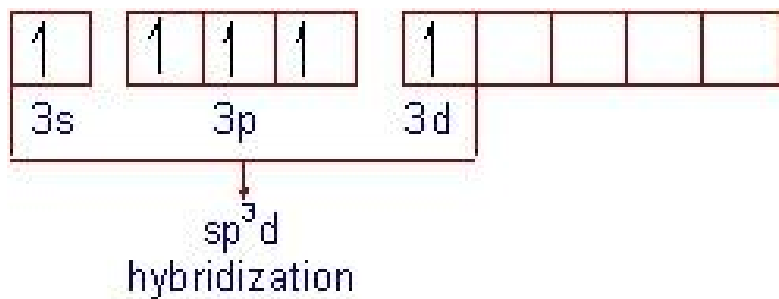


➤4) Phosphorus pentachloride(PCl_5)

The ground state electronic configuration of phosphorus atom is: $1s^2 2s^2 2p^6 3s^2 3p_x^1 3p_y^1 3p_z^1$.

Thus the electronic configuration of 'P' in the excited state is $1s^2 2s^2 2p^6 3s^1 3p_x^1 3p_y^1 3p_z^1 3d^1$.

In the excited state, intermixing of a 3s, three 3p and one 3d orbitals to give five half filled sp^3d hybrid orbitals, which are arranged in trigonal bipyramidal symmetry



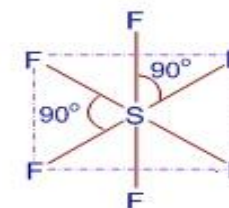
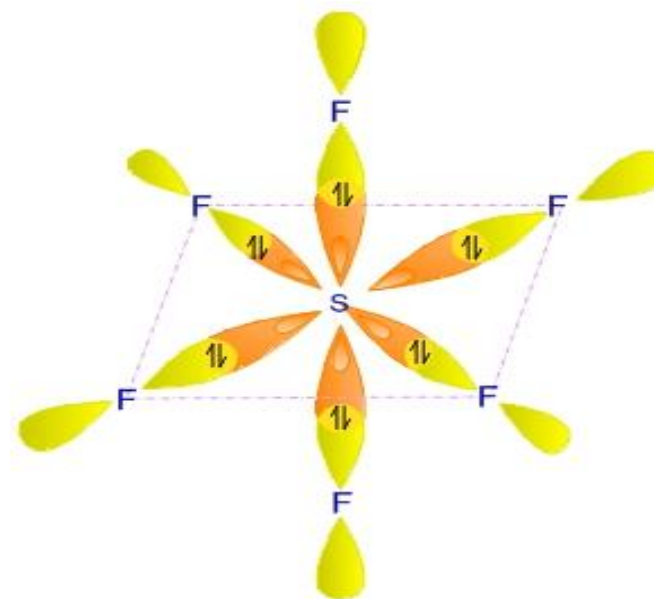
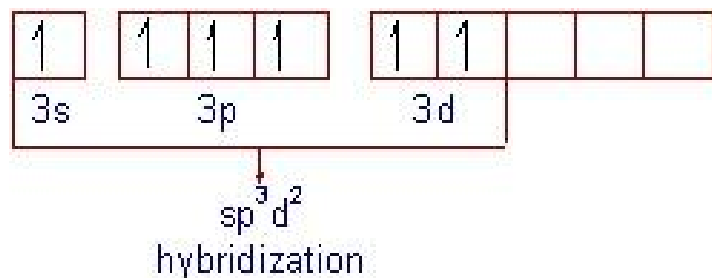
Trigonal bipyramidal structure of PCl_5

➤ 5) Sulfur hexafluoride (SF₆)

The electronic configuration of 'S' in ground state is $1s^2 2s^2 2p^6 3s^2 3p_x^2 3p_y^1 3p_z^1$.

Thus the electronic configuration of 'S' in its 2nd excited state is $1s^2 2s^2 2p^6 3s^1 3p_x^1 3p_y^1 3p_z^1 3d^2$.

In the second excited state, sulfur undergoes sp^3d^2 hybridization by mixing a 3s, three 3p and two 3d orbitals. Thus formed six half filled sp^3d^2 hybrid orbitals are arranged in octahedral symmetry



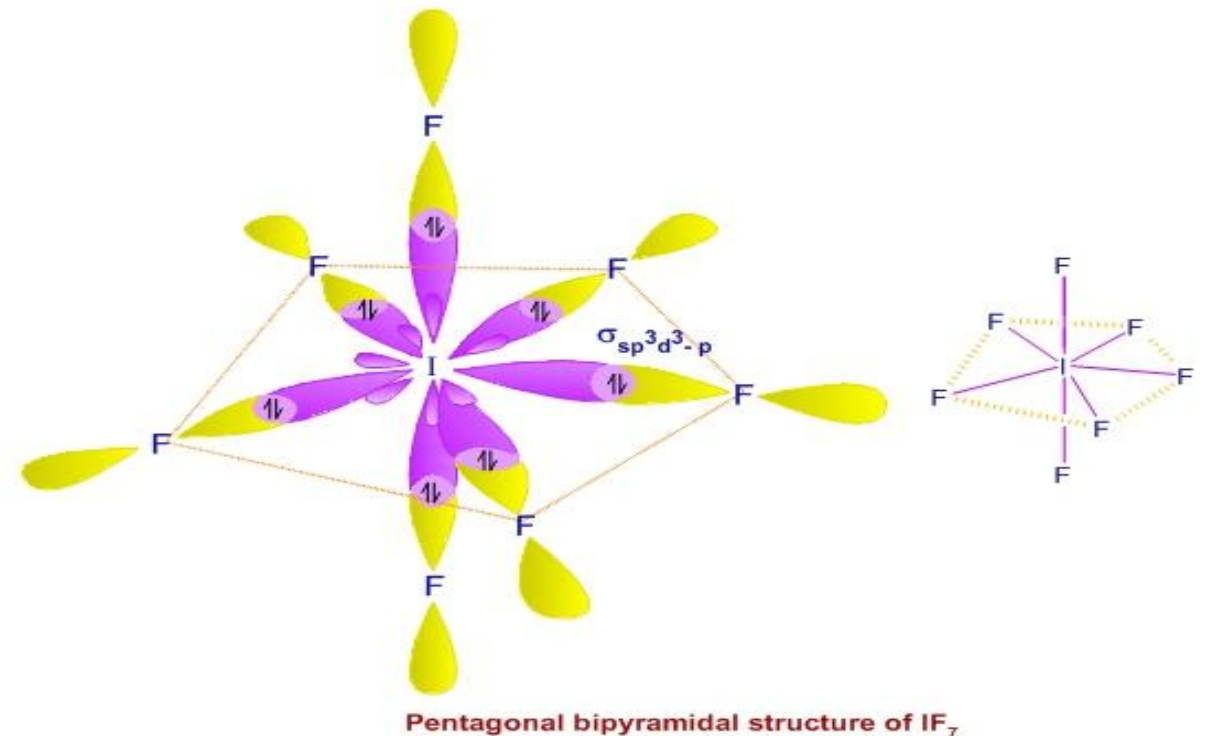
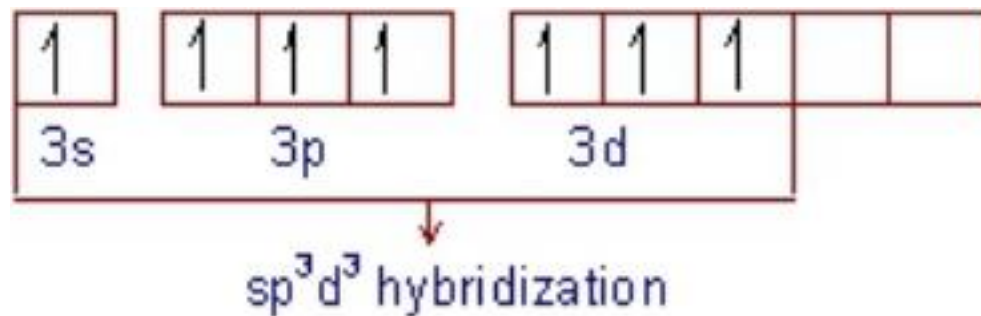
octahedral structure of sulfur hexafluoride

➤ 6) Iodine heptafluoride (IF₇):

The electronic configuration of Iodine atom in the ground state is: [Kr]4d¹⁰5s²5p⁵.

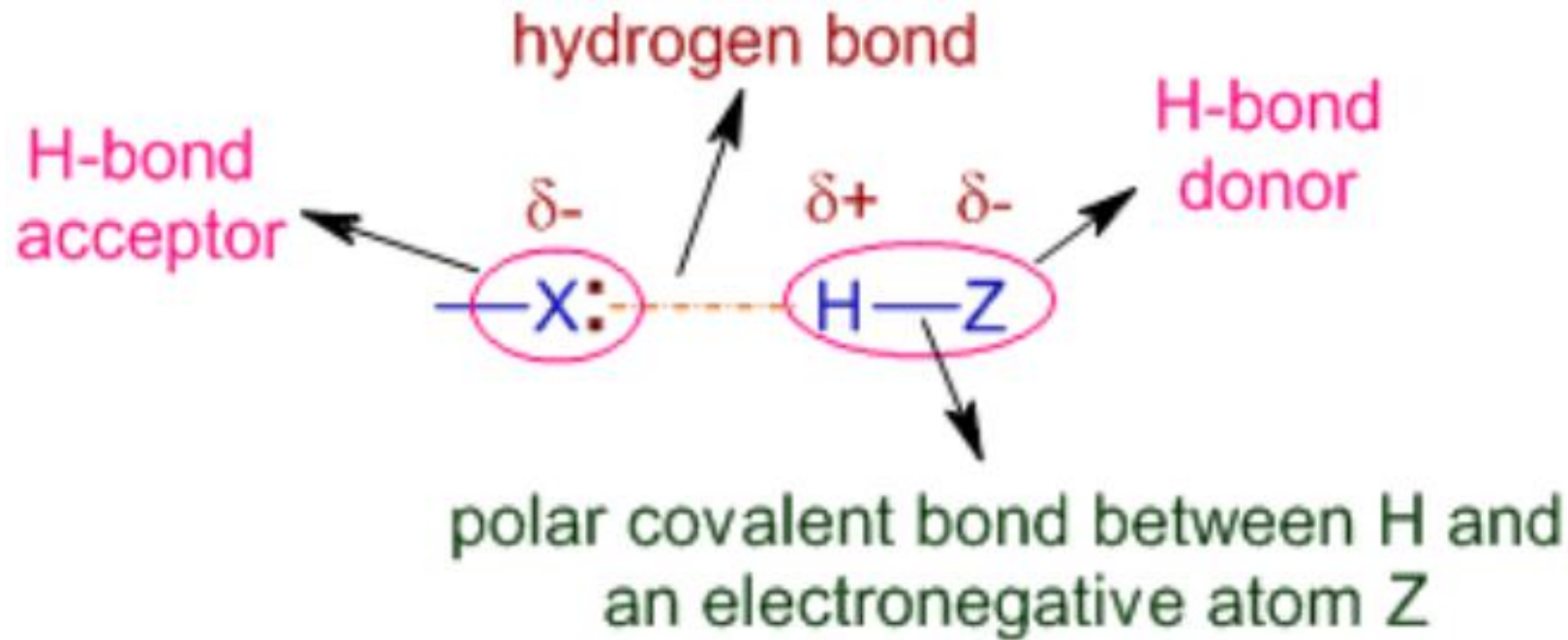
The electronic configuration of Iodine in the third excited state can be written as: [Kr]4d¹⁰5s¹5p³5d³.

In the third excited state, iodine atom undergoes sp³d³ hybridization to give 7 half filled sp³d³ hybrid orbitals in pentagonal bipyramidal symmetry. These will form 7 σsp³d³- p bonds with fluorine atoms.



Hydrogen Bond

The electrostatic attraction between partially positively charged hydrogen atom bonded to a relatively electronegative atom and another electronegative atom is referred to as hydrogen bond.



Characteristics of Hydrogen Bond

1) The hydrogen atom participating in H-bond must be covalently bonded to a highly electronegative atom in order to acquire significant positive charge.

For example, H atom covalently bonded to electronegative atoms like F, O and N can participate in hydrogen bonding.

However, H atoms bonded to less electronegative atoms like B, C etc. or larger electronegative atoms like S, P, Cl, Br etc. cannot get sufficient positive charge and hence cannot enter into H-bonding. Nevertheless there are some special cases.

2) The acceptor electronegative atom should also possess at least one lone pair and high charge density so as to attract positively charged H atom effectively. Thus H-bonding is normally observed with smaller and highly electronegative atoms such as F, O and N with lone pairs.

3) The H-bond is mostly electrostatic in nature. However, directional nature of hydrogen bonding, shorter inter atomic distances between molecules participating in H-bonding and limited number of H-bond interactions suggest some covalent nature also.

E.g. In HF, the number of H-bonds per each molecule is limited to only two, since there is only one hydrogen per molecule and only one lone pair on Fluorine can participate in H-bonding. The directional nature is also evident from the following diagram.

4) The hydrogen bond is stronger than van der Waal's forces of attraction. But it is weaker than the covalent bond and ionic bond.

The strength of Hydrogen bonds can range from very weak ($1\text{--}2\text{ kJ mol}^{-1}$) to extremely strong (161.5 kJ mol^{-1}). H-bonds involving fluorine are associated with higher energies. Most of the remaining types are below the range of $21\text{--}25\text{ kJ mol}^{-1}$.

5) The strength of H-bond is directly proportional to the electronegativity and the ability of electronegative acceptor atom to contribute the electron density.

H-bond	Energy in vapor state
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F-H----F	161.5 kJ mol^{-1}
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O-H----O	21 kJ mol^{-1}
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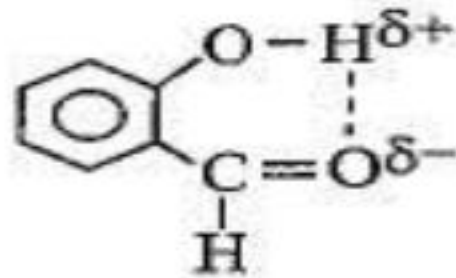
N-H----N	13 kJ mol^{-1}
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O-H----N	29 kJ mol^{-1}
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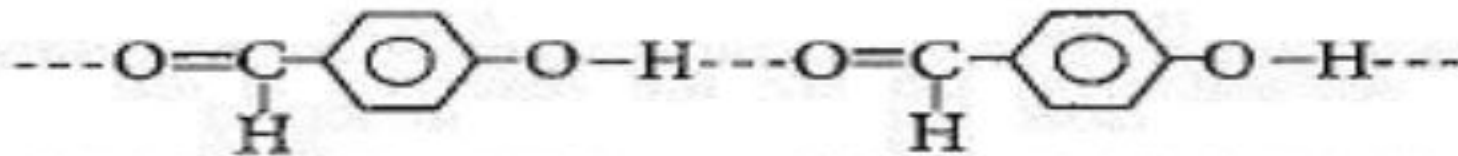
N-H----O	8 kJ mol^{-1}
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Intermolecular H-bond

- The H-bond formed between two different molecules (of same or different compounds) is referred to as intermolecular H-bond. In this case, hydrogen of one molecule makes H-bond with electronegative atom of another molecule. These molecules may belong to same compound or different compounds.



o-hydroxybenzaldehyde
(intramolecular hydrogen bonding)



p-hydroxybenzaldehyde

(intermolecular hydrogen bonding)

Consequences of intermolecular hydrogen bonding

1) The boiling point, melting point, sublimation point as well as enthalpy of vaporization of compounds increase due to intermolecular H-bonding.

E.g. The boiling point of H_2O is high (373 K) due to intermolecular hydrogen bonding and hence it exist as liquid at room temperature, whereas H-bonds are absent in H_2S and hence it has low boiling point and exists as a gas at room temperature because the electronegativity of sulphur is low compared to oxygen so, the H bonding is negligible in case of H_2S and the only binding force is weak Van Der Waals attraction between the molecules.

Note: Though sulfur is an electronegative element, it cannot form hydrogen bonds due to larger size. There is no ample charge density on either sulfur or hydrogen to make a H-bond.

2) The solubility of compounds, showing intermolecular H-bonding, in water increases due to formation of intermolecular H-bonds with water.

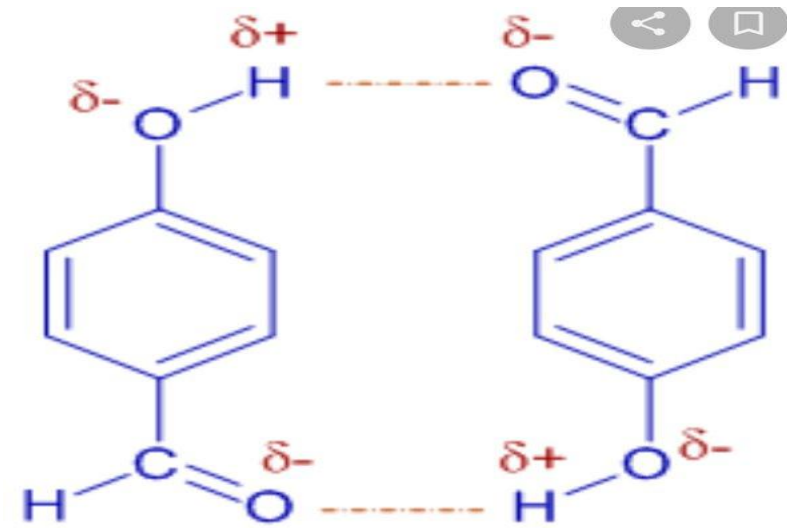
E.g. Ethyl alcohol, methyl alcohol, ammonia, HF, acetic acid etc., are fairly soluble in water due to their ability to form intermolecular hydrogen bonds with water molecules.

Due to intermolecular H-bonding, para isomers have higher boiling points and are fairly soluble in water. Therefore they cannot be separated by steam distillation.

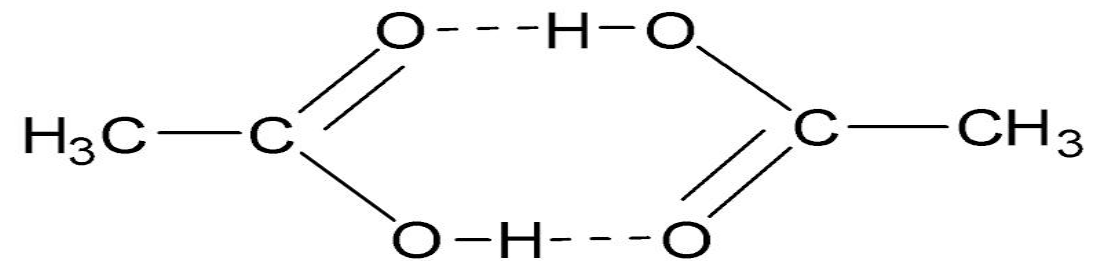
E.g. Para hydroxy benzaldehyde has high boiling point and is more soluble in water, than its ortho isomer, due to intermolecular H-bonding and hence cannot be separated by steam distillation.

4) Some molecules exist as dimers or trimers or polymers in non-aqueous medium or in vapor state due to strong intermolecular H-bonding.

E.g. Acetic acid exists as dimer in benzene.



Intermolecular hydrogen bonding in para hydroxy benzaldehyde



Dimerization of acetic acid

Intra-molecular H-bond

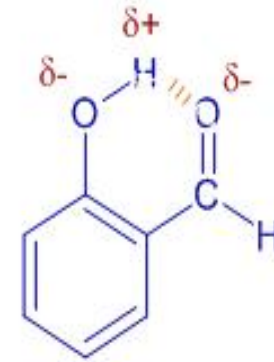
The H-bond formed between a hydrogen and the electronegative atom of same molecule is called as intra-molecular H-bond. This is also termed as intra-molecular chelation, since a ring is formed within the molecule.

Consequences of intra-molecular hydrogen bonding

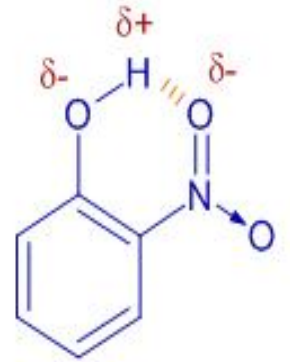
1) The boiling point, melting point and solubilities of isomers containing intra-molecular hydrogen bonds are lower than their isomers showing intermolecular H- bonds.

E.g. The boiling point and solubility of ortho hydroxy benzaldehyde (salicylaldehyde) are lower than those of para hydroxy benzaldehyde.

In ortho hydroxy benzaldehyde, there are intra-molecular hydrogen bonds, whereas para hydroxy benzaldehyde forms inter molecular H-bonds. Hence ortho hydroxy benzaldehyde is steam volatile and can be separated easily from other isomers by steam distillation.



Intramolecular hydrogen bonding in ortho hydroxy benzaldehyde (salicylaldehyde)



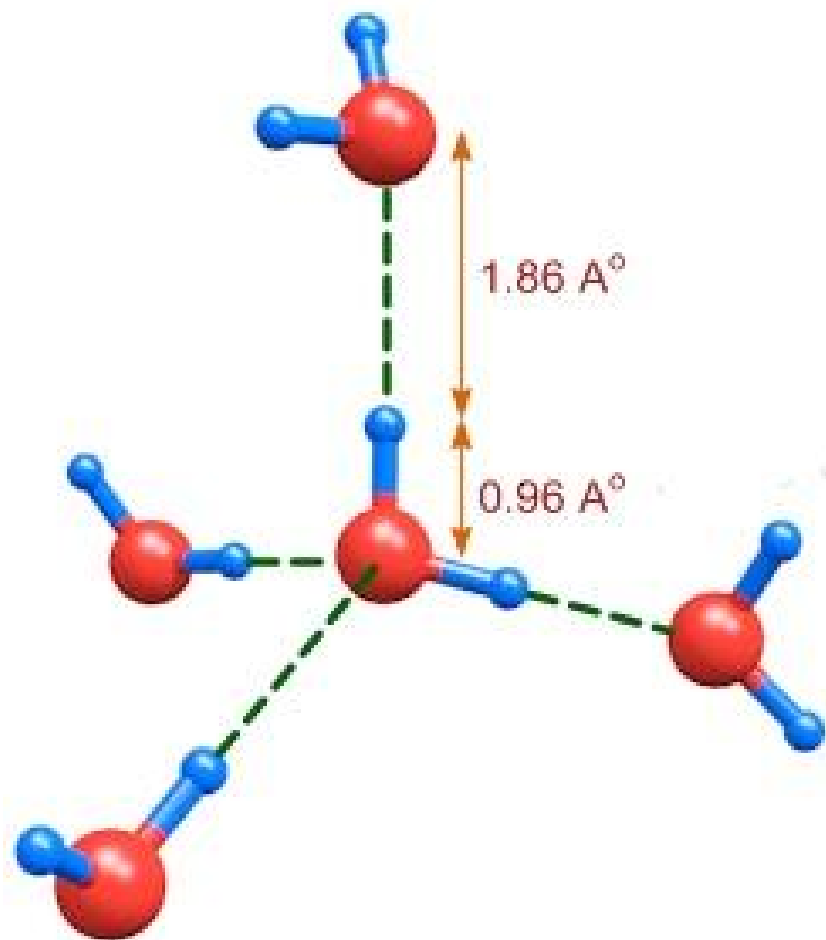
Intramolecular hydrogen bonding in ortho nitrophenol

In general, boiling point, melting point and solubility in water are lower for ortho isomers due to intra-molecular hydrogen bonding when compared to their para isomers.

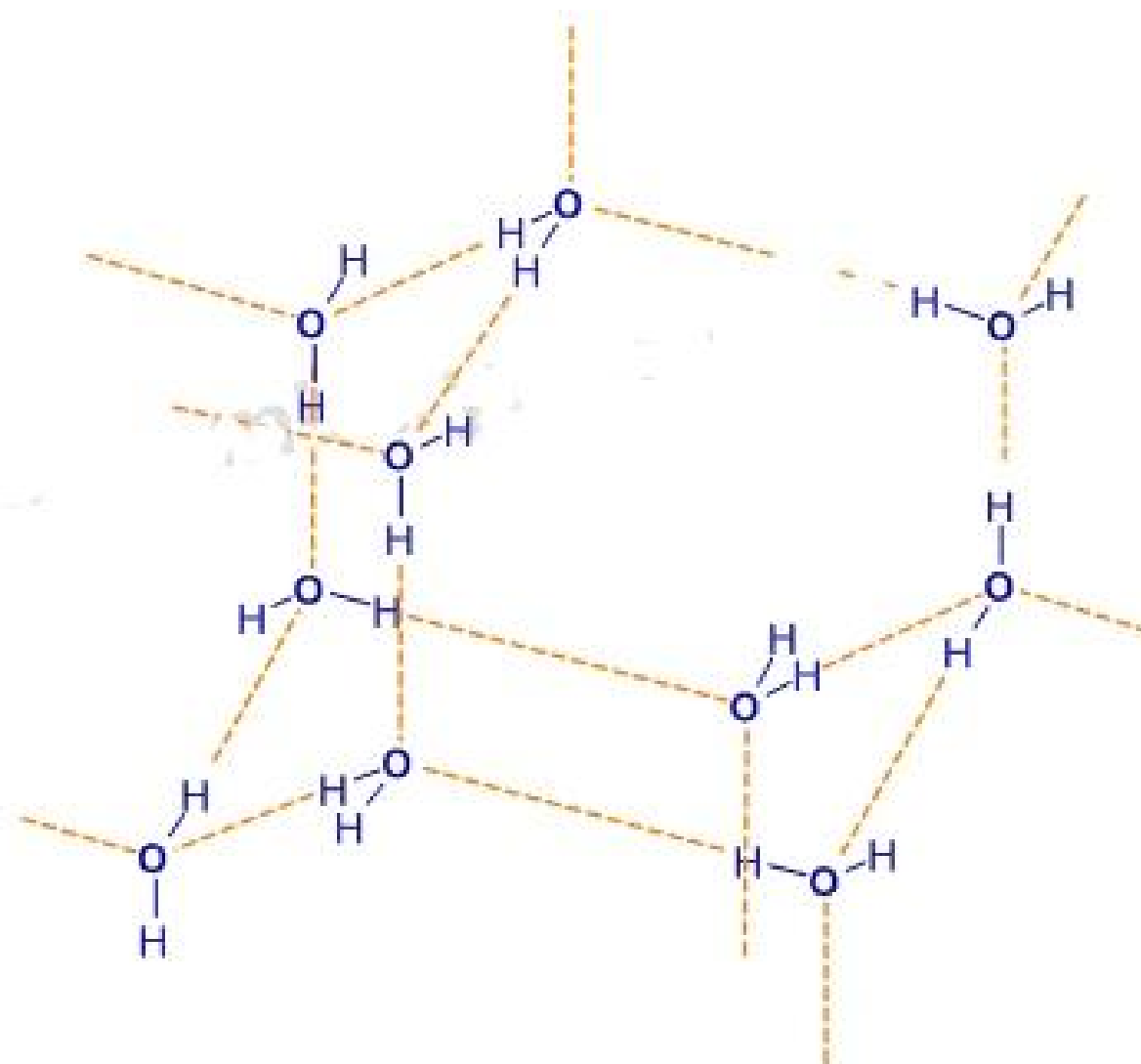
Density of ice is less than water

Solid water (ice) has less density than water due to stable hydrogen bonding which forces the crystalline structure to be very open (hexagonal type of) creating some extra interstitial space. Hence ice occupies more volume and hence less density than water.

In water, the two hydrogen bonds and one oxygen bond are joined with each other by a polar covalent bond. The water molecule is non-linear structure and has a bent geometry and due to the electronegative difference between the oxygen atom and hydrogen atom, partial negative charge is generated on the oxygen and partial positive charge is generated in the hydrogen. Due to this, the water molecule attracts other water molecules to form a hydrogen bond between them. In the liquid form, the hydrogen bond is continuously formed and broken due to the movement of water molecules in the system due to the presence of heat. When the temperature has increased the breaking of the bond is increased with the increased movement of a water molecule. When the temperature is reduced and water is frozen and forms a crystallized structure made up of hydrogen bonds. A very high amount of energy is required to break the bond in the crystalline structure. In the solid-state of water, each oxygen atom is surrounded by four hydrogen atoms which are loosely held. The two hydrogens are covalently bonded to the oxygen atom and two other hydrogen atoms are bonded to the lone pairs of the oxygen atoms. Due to the open porous structure of ice, the density of ice is less than water.



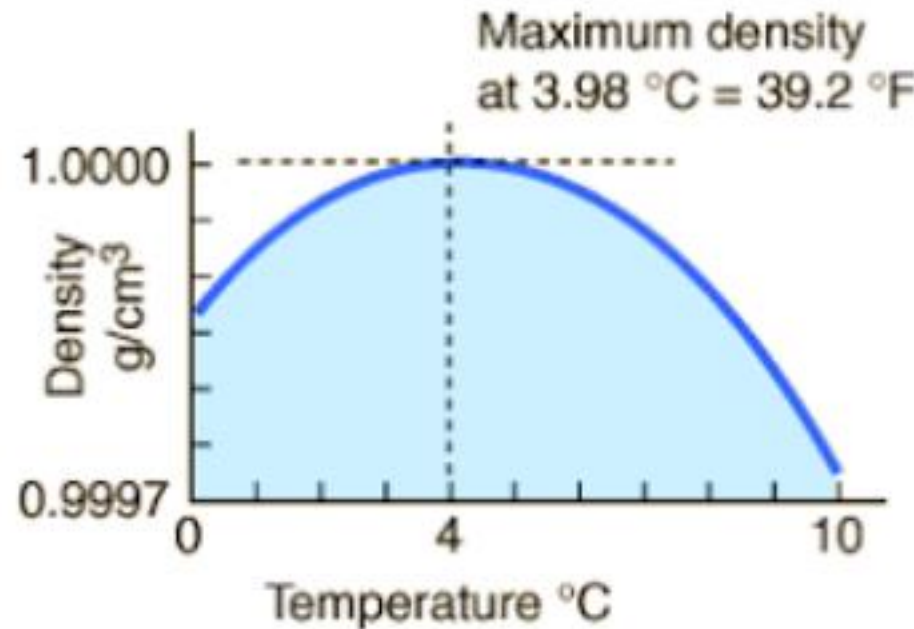
Hydrogen bonding in water
 Red = oxygen
 Blue = hydrogen



crystal structure of ice showing H-bond interactions

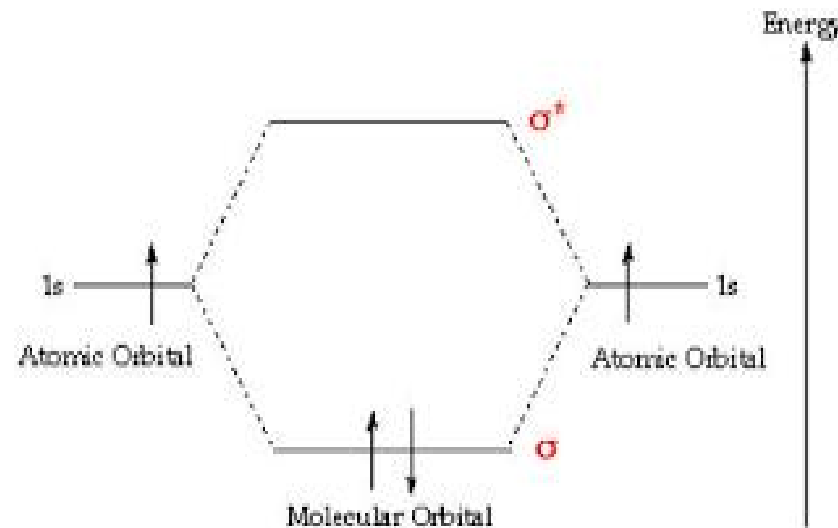
Density of water is maximum at 4°C

As the temperature of liquid water (at 0°C) is increased above 4°C, density of water also increases. The increase in density is continuous only up to 4°C. Above 4°C, kinetic energy of H₂O molecules increases and hence H₂O molecules move away from each other, i.e. the volume of water molecules is increased. The increase in volume decreases the density of water. Therefore above 4°C, the density of water starts decreasing again. Thus we see that the density of water is maximum at 4°C.

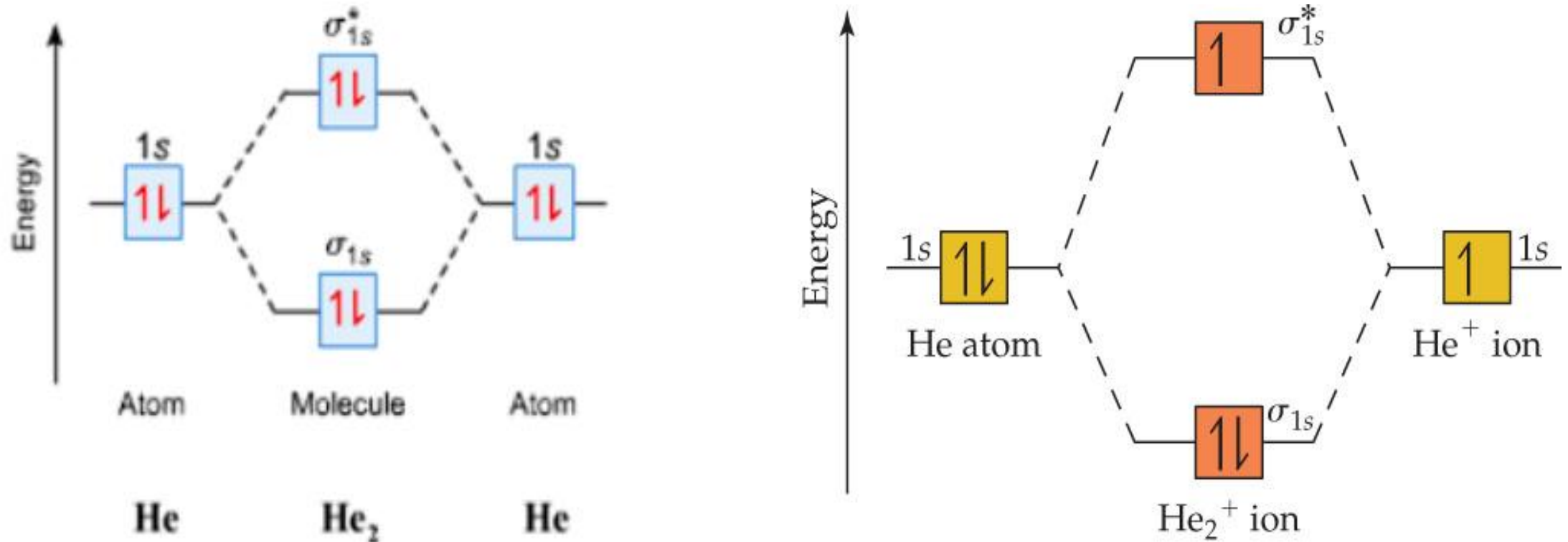


Molecular Orbital Theory

- The number of molecular orbitals produced is always equal to the number of atomic orbitals brought by the atoms that have combined.
- Bonding molecular orbitals are lower in energy than the parent orbitals, and the antibonding orbitals are higher in energy.
- Electrons of the molecule are assigned to orbitals from lowest to successively higher energy.
- Atomic orbitals combine to form molecular orbitals most effectively when the atomic orbitals are of similar energy.



He_2^+ exists but He_2 does not

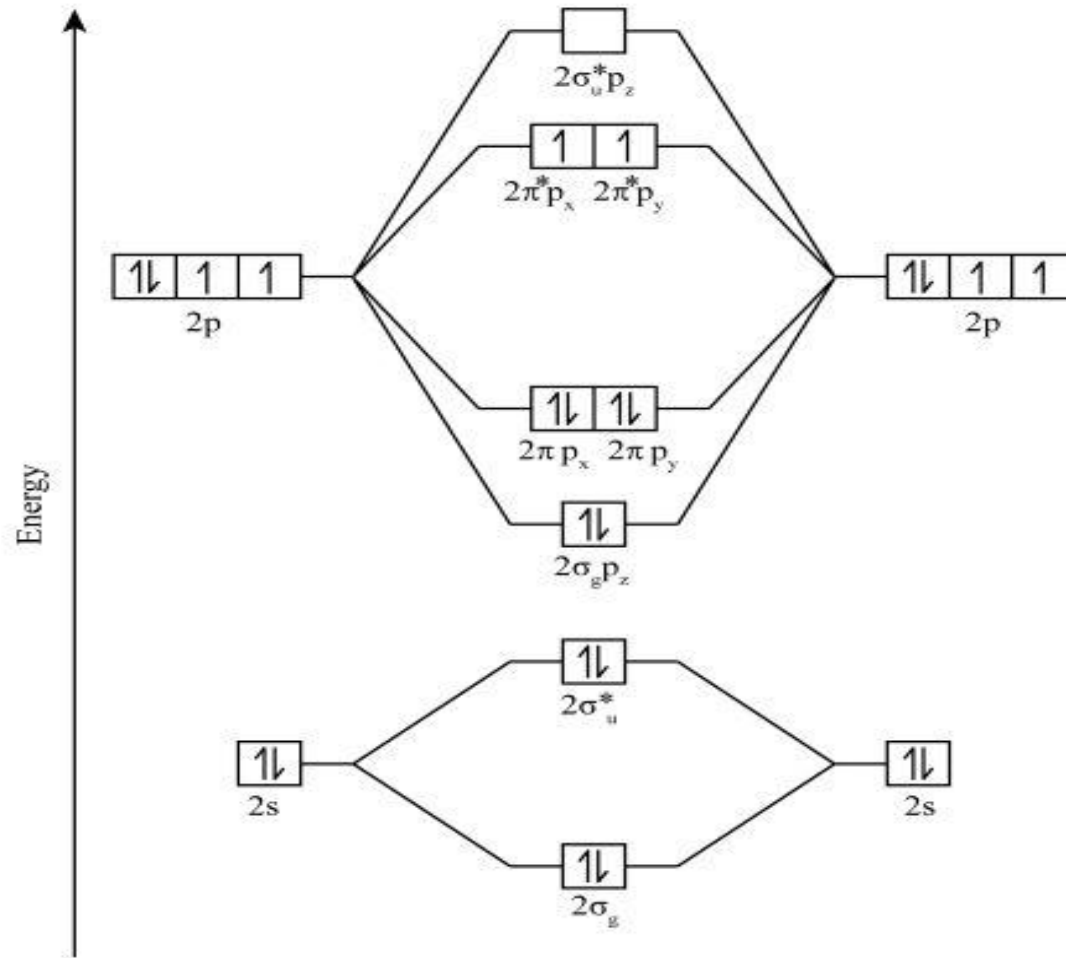


Bond order = $\frac{1}{2}(\text{No. of bonding electrons} - \text{No. of Anti bonding electrons})$

If we calculate the bond order of He_2 molecule it comes out to be 0 .

But bond order for He_2^+ comes out to be 0.5 so it exists

O_2 is paramagnetic



Its B.O. is $\frac{1}{2} (8-4)=2$. Thus O_2 consists of double bond. It is paramagnetic due to presence of two unpaired electrons.